A-A-54305
1 April 1991

COMMERCIAL ITEM DESCRIPTION

ETHYLENE OXIDE-DICHLORODIFLUOROMETHANE MIXTURE, Filled Cylinder

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a gas mixture of ethylene oxide and dichlorodifluoromethane suitable for use as a sterilant.

Salient characteristics:

General. The mixture shall be a gas consisting of ethylene oxide and dichlorodifluoromethane in accordance with the requirements contained herein. The mixture shall be approximately 135 pounds of non-flammable gas and shall contain not less than 11.25 percent and not more than 12.25 percent (by weight) of ethylene oxide with the remainder consisting of dichlorodifluoromethane. The mixture shall be used as a sterilant with certain gas sterilizers (See Note 2).

Volume. The specific volume of the gas mixture shall be 3.766 cubic feet per pound, when measured at 70°F and 1 atmosphere of pressure. Each aliquot shall yield the same percentage mixture.

Ethylene Oxide. The liquid ethylene oxide (active ingredient) shall be substantially free of suspended material and shall comply with the following requirements:

Flash point (open cup test)	- 0°F.
Water solubility	- Complete
Acidity (& by weight)	- 0.002 maximum
Water (% by weight)	- 0.03 maximum
Aldehydes (% by weight) acetaldehyde)	- 0.01 maximum (as
Acetylene	- None
Nonvolatiles	- 0.010 gram/100 ml maximum
Odor	- Nonresidual

AMSC N/A FSC 6505

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Dichlorodifluoromethane. The liquid dichlorodifluoromethane shall be substantially free of suspended material and comply with the following requirements:

Water solubility (by weight)	- 0.008 at 70°F.
Water (% by weight)	- 10 p.p.m. maximum
Odor	- Nonresidual
Nonvolatiles	- 0.01 gram/100 ml maximum
Presence of chloride as HCl	- None

Gas Cylinder. The cylinder for the ethylene oxide dichlorodifluoromethane mixture shall be in accordance with the following requirements:

The cylinder shall be a non-shatterable, impact-resistant, compressed gas cylinder, suitable for containing 14 gallons of ethylene oxide-dichlorodifluoromethane mixture for use in conjunction with certain gas sterilizers (See Note 2).

The cylinder shall be equipped with American National Standards Institute (ANSI) Connection No. 510 as indicated in ANSI Standard B57.1. The valve shall have a chromium plated finish.

Cylinder dimensions shall be approximately 12 inches in diameter by 38 inches in height and the cylinder shall be oil-free.

The cylinder shall be provided with a neck-ring and valve protective cap which may be perforated.

The cylinder shall comply with the requirements of Department of Transportation Standard 3AA for compressed gas cylinders.

Unless otherwise specified, the cylinder shall be furnished by the Government for filling with the ethylene oxide-dichlorodifluoromethane gas mixture.

Labeling. Labeling of the cylinder shall be in accordance with applicable regulations.

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Stamping, stenciling and marking. At the time of filling, the cylinders shall be stamped with the gas symbol "C_2H_4O and CCL_2F_2" and stenciled with the name "ETHYLENE OXIDE DICHLORODIFLUOROMETHANE MIXTURE". Markings shall be not less than 1/4 inch high and shall be plainly and permanently marked on the shoulder of each cylinder starting near the neck. Each filled cylinder shall be marked with the legend: "STORE IN A COOL PLACE (56° to 80°)".

Tagging. A special tag constructed in accordance with commercial practice shall be attached to each filled cylinder. Marking of this tag shall include the name of the gas, as specified in the contract, serial number of the cylinder, and necessary precautionary markings.

Tags shall be furnished with a metal eyelet and stringing wire. The wire shall be tinned, annealed, or galvanized, and be 23 gage or greater in diameter.

Tag colors as follows:

Background - Same color as cylinders
Print - Black

or

Background White
Print - Same color as cylinders

In addition, the bottom of the tag shall be designed with a three part legend, marked on both sides of the tag, as follows:

"EMPTY"
"IN USE"
"FULL"

A perforation shall be placed between the word "FULL" and "IN USE" and between the words "IN USE" and "EMPTY" in order that these may be removed by tearing, when they are no longer applicable.

Unit. Each (EA). One filled cylinder, as specified, constitutes one unit.

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QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more text characteristics.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as specified herein. Unit of product for examination purposes shall be one cylinder. Sampling shall be inspection level II with an acceptable quality level (AQL) of 1.0.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for test purposes shall be one cylinder. Sampling shall be inspection level S-2 with an AQL of 1.0.

Tests. Tests shall be conducted to determined compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers'own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and /or order.

NOTE 1: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6505-00-735-7194	ETHYLENE OXIDE-DICHLORODIFLUOROMETHANE Mixture, Filled Cylinder, 135 lb (61.25 Kg)

NOTE 2: The two types of sterilizers used in conjunction with the ethylene oxide-dichlorodifluoromethane mixture are covered by National Stock Numbers (NSNs) 6530-00-890-1976 (Sterilizer, Heat and Moisture Labile Medical Instruments and Supplies) and NSN 6530-00-735-7195 (Sterilizer, Surgical Instruments and Dressing).

Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6505-5365

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54305	2. DOCUMENT DATE (YYMMDD) 1 April 1991

3. DOCUMENT TITLE
ETHYLENE OXIDE-DICHLORODIFLUROMETHANE MIXTURE, Filled Cylinder

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center	(215)737-2870	444-2870
c. ADDRESS (Include Zip Code) ATTN: DPSC-MSTE 2800 S. 20th Street Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — Previous editions are obsolete — 90/790

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